

# Preliminary analyses on technical and economic viability of moon-mined propellant for on-orbit refueling

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## ABSTRACT

The present study analyses the technical feasibility and economic sustainability of Moon-mined propellant transportation, storage, and distribution to final users. Two technical architectural options are considered for propellant transportation: taking fuel from the Moon or from Earth. Both architectures assume the same storage and distribution system, i.e., the use of an orbiting depot. The depot will be reachable by various types of spacecrafts for refueling. Both architectures are evaluated in terms of technical feasibility, performance and economic convenience. As the estimates of the investment, operational costs, and revenues are highly uncertain, a risk evaluation, i.e. the Monte Carlo analysis, is used to evaluate the exposure of the ventures to factors that could affect the volatility of results and lead them to fail. These analyses identify transporting propellant from the Moon as the most promising solution.

## 1. Introduction

Several science missions (e.g. the SOFIA Mission of NASA 2020, among others [1]) provided the evidence of high concentration of water on Moon poles. This abundance of resources paves the way for the expansion of space activities and the development of new business models. Previous works have researched and confirmed the technical feasibility of building a mining plant on the Moon surface to provide water extraction and derive propellant from it [2]. The prospect of propellant availability in space would affect many sectors of space economy while enhancing deep space exploration.

On-orbit refueling is performed on a space-based facility (client) by a specifically designed space vehicle (supplier). There are several benefits associated to it, which range from the extension of a mission's lifetime and the improvement of its flexibility – with regards to mission costs reduction and payload mass increase – to the opportunity of enabling new missions by reducing the difficulty of travelling beyond Low Earth Orbit (LEO). This can lead to new business models for space activities and enormous potential market implications such as providing demand

for materials mined from the Moon and, in the long-term, from asteroids. In particular, the availability of propellant produced on the Moon and supplied to clients by a distribution architecture is key for the development of cis-lunar space activity and beyond. By leveraging lunar ice, rockets could potentially refuel once in space, thus being allowed to reach distant locations. Since the amount of energy required for ascent and descent maneuvers on the Moon surface is considerably less than that needed to escape Earth's gravity, the propellant distribution activities become less expensive in terms of propellant consumption. Despite the realization of such architecture for cryogenic propellant transport, storage and distribution in cis-lunar space demands to face nontrivial technical challenges, the potential scientific benefits and positive market implications outweigh the disadvantages.

The objective of the present study, result of a collaboration between the Mechanical and Aerospace Engineering Department (DIMA) of Sapienza University of Rome and SDA Bocconi School of Management, is to understand the economics behind an infrastructure that can enable the refueling of spacecrafts with lunar propellant. Building on the results of a previous research addressing the economic viability of water

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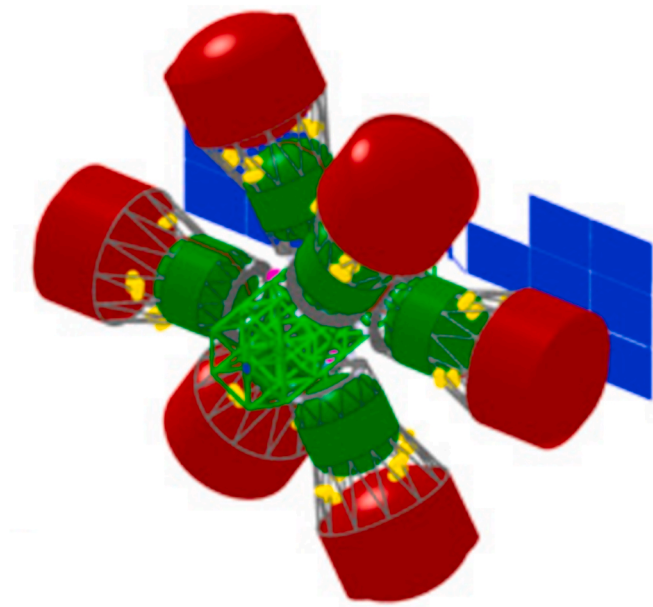


Fig. 1. Example of LOX/LH2 depots in assembled configuration [5].

extraction on the Moon surface to produce in-situ liquid hydrogen (LH2) and oxygen (LOX), this study evaluates both the technical feasibility and the economical convenience of a system for propellant transportation, storage, and distribution, which can provide on-orbit refueling to potential users.

## 2. Adopted approach

At first, the technical viability analysis is conducted to determine the optimal configuration of the propellant distribution facility by examining the engineering feasibility of the infrastructure and evaluating the engineering costs related to it. The analysis, based upon the assumption that a mining plant on the moon surface is technically feasible and able to provide cryogenic propellant from the moon's ice, is performed considering state of the art of on-orbit refueling and available engineering solutions in terms of technologies for cryogenic propellant storage and transfer. As a result, a depot-centric propellant distribution architecture is determined together with its optimal placement at Lagrangian point L1. The depot is envisioned to be served by a vehicle (here and henceforth referred to as “Moon shuttle”) which performs ascent and descent maneuvers from/to the Moon surface and L1 to refuel the depot's tanks with Moon-mined propellant. This approach is compared with the option of servicing the depot by a heavy launch vehicle, which takes propellant from the Earth. Once full, the depot is intended to be available for propellant transfer to clients performing refueling docking maneuvers.

The definition of the propellant distribution architecture features various steps: (i) the selection of the type of distribution architecture to be considered, (ii) the determination the of optimal placement of the distribution facility, (iii) a preliminary design of the distribution facility, (iv) the analysis of refueling maneuvers associated to the distribution architecture and finally (v) an estimation of the overall costs associated to the distribution pattern, considering not only the initial necessary investment, but also the operational costs, maintenance, and expansion costs.

With the technical infrastructure and an estimation of the overall investment costs, it is then possible to analyze the economic convenience. We assume that we are in a market composed by a mining company, a monopolistic transporter/distributor company, and end-users. The mining company only sells fuel on the Moon at a fixed

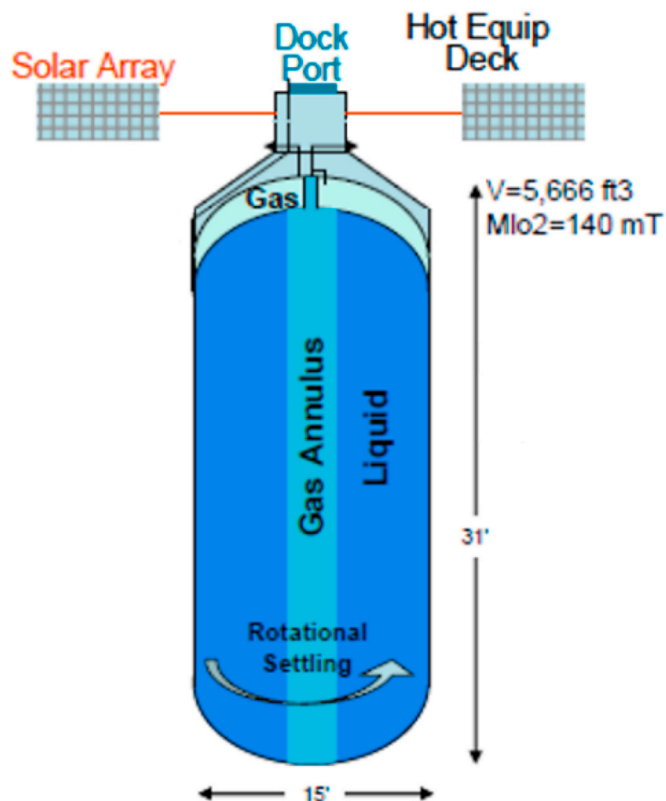


Fig. 2. Example of a fluid's management system (spin around longitudinal axis) [5].

price of 500 \$/kg as in Refs. [1,2]. The monopolistic transporter/distributor company buys propellant from the Moon and transports it to the deposit in L1.

This approach is compared with a transporter which buys propellant from Earth and transports it to the deposit. We will see that transporting fuel from the Moon needs a high initial investment to research and develop the Moon Shuttle, on the other hand transporting fuel from Earth will have higher operational costs caused by Earth's gravity. Through the Net Present Value (NPV) analyses, it is possible to weigh costs and benefits of these approaches.

The viability of the two projects is strongly affected by the potential selling price of the propellant, a value that is determined from the ratio of the net present value of the expected operating cash flows and the cost of the initial capital of the suppliers.

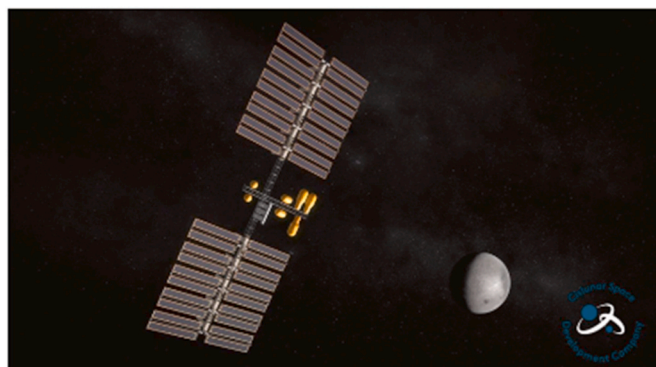


Fig. 3. Lunar propellant depot in L1 [9].

**Table 1**  
The characteristics of the designed OD.

| LOX mass [t]                        | LH2 mass [t] | Depot dry mass [t] | Material | Dimensions [m]                    |
|-------------------------------------|--------------|--------------------|----------|-----------------------------------|
| 21.43                               | 3.57         | 3.47               | SS 301   | Height h = 5.79<br>Diameter d = 4 |
| Depot's subsystems mass budget [kg] |              |                    |          |                                   |
| Propulsion                          |              |                    |          | 927.73                            |
| AOCS                                |              |                    |          | 173.95                            |
| EPS                                 |              |                    |          | 1,043.70                          |
| TCS                                 |              |                    |          | 139.16                            |
| DMS                                 |              |                    |          | 69.58                             |
| TTC                                 |              |                    |          | 34.79                             |
| Structure                           |              |                    |          | 521.85                            |
| Harness                             |              |                    |          | 104.37                            |
| Dry Mass (with margin)              |              |                    |          | 3479                              |

### 3. Technical viability

#### 3.1. Selection of the distribution architecture

Improving a space-based capability through a combination of on-orbit activities is the most viable path forward for continuing to expand space activities beyond their present limitations and enable new missions. The process of performing servicing operation in space such as inspection, orbit modification and maintenance, upgrade and repair, assembly, debris mitigation and refueling is known as On-Orbit Servicing (OOS). On-orbit refueling is a servicing operation attainable using either Orbiting Depots (OD) [3] or a distributed launch architecture resorting to Heavy Lift (HL) launches [4].

The HL configuration is limited by the performances of current launchers in terms of payload capability and  $\Delta V$  requirements [4]. The technique is based on an in-space assembly of upper stages with the main purpose of performing an ad-hoc propellant transfer to a vehicle. This approach is subject to higher failure probability because of the multi-launch architecture involving multiple systems. Additionally, the



Fig. 4. CSDC's Moon shuttle rendering [20].

**Table 2**  
CSDC's Moon shuttle specs [20].

| Moon Shuttle – CSDC    |                      |                      |         |                                    |
|------------------------|----------------------|----------------------|---------|------------------------------------|
| Payload capability [t] | Vehicle dry mass [t] | Tanks capability [t] | Isp [s] | Dimensions [m]                     |
| 25                     | 14.2                 | 69                   | 450     | Height h = 8.1<br>Diameter d = 7.5 |

system is specific for each space mission: the payload, the launch windows and the on-orbit docking are in fact designed to support a particular end-user scenario [4]. ODs represent a far less mission-dependent option when considering possible propellant distribution architectures. In this case the depot servicing platforms are located where the propellant is stored and transferred to vehicles as needed so that the mission and depot's design are meant to serve various types of users. This configures a flexible and adaptable system that can be easily launched in an assembled configuration [5] (see Fig. 1). Both HL and OD options require storing the propellant, which usually consists of liquid oxygen and hydrogen, in cryogenic conditions. The HL approach requires less storage time than the OD architecture as the latter is meant to represent a flexible solution for any kind of space mission. On the contrary, in the HL configuration the upper stage should be equipped by all the subsystems required to store hydrogen and oxygen in liquid phase for each launch, increasing the mass amount [6].

Based on these considerations, the depot-centric configuration is selected as the most suitable option to enable on-orbit refueling in cis-lunar space. The depot should be designed as a space vehicle equipped with propellant storage tanks placed in a suitable point in space, for example LEO orbit or Lagrangian point L1.

#### 3.2. Optimal placement of the orbiting depot

The placement of the distribution facility must consider the current and forecast in-space market demand [7]. To this purpose, both LEO, low lunar orbit (LLO) and the Lagrangian point L1 are strategic selling points for a beyond-Earth In-Situ Resource Utilization (ISRU) facility to support space missions and exploration [2]. A trade-off analysis between different possible OD placements is performed considering four main factors, such as technical requirements related to cryogenic propellant storage and transfer, launch windows to reach both the depot's placement and the client's destination, and potential market demand.

On the technical requirements side, cryogenic propellants require complex systems for fluid management and thermal control to minimize the propellant boil-off driven by heat transfer that could hinder cryogenic conditions (see Fig. 2). Existing technologies and materials are considered capable of guaranteeing the necessary thermal management performances [5]: in this sense, among the possible OD placement options, LEO and LLO orbits are particularly unfavorable due to the wide temperature range experienced caused by, among the others, solar radiation and terrestrial albedo [8]. The Lagrangian point L1 appears instead to be a more suitable solution [8] (see Fig. 3).

One of the most important benefits of L1 placement is related to launch windows, which are continuously open both to/from Earth and to/from the lunar surface [10]. This would facilitate refueling maneuvers from the lunar plant by making them always possible and it would

**Table 3**  
OD and CSDC's Moon shuttle DDT&E costs [22].

| Vehicles DDT&E and Manufacturing Costs [M\$] |       |
|----------------------------------------------|-------|
| Depot DDT&E Costs                            | 400   |
| Depot Manufacturing Costs                    | 25    |
| Moon Shuttle DDT&E Costs                     | 1,200 |
| Moon Shuttle Manufacturing Costs             | 700   |

**Table 4**

CSDC's Moon shuttle transportation cost via NASA's SLS [22].

| Moon Shuttle transportation on Moon – NASA's SLS launcher |        |
|-----------------------------------------------------------|--------|
| Lift off mass [t]                                         | 14.2   |
| Launch cost [\$/kg]                                       | 47,183 |
| Total launch cost [M\$]                                   | 670    |

support lunar missions being a direct link between the Earth and the Moon. Reaching the Moon surface is possible from LEO too, even though the launch window is every 9 days. However, considering potential users' destinations, interplanetary departure from LEO is not guaranteed during an interplanetary transfer window due to the orbit precession of the depot over time, while it is guaranteed from L1 [11]. Despite the convenience of the launch windows, a drawback of L1 placement is the expensive braking maneuver which has to be performed to dock when arriving from LEO [12]. When considering refueling maneuvers from the Moon surface, on the other hand, L1 becomes particularly advantageous [6]: from here, it would also be possible to refuel a LEO depot with lunar propellant through a low-cost maneuver which resorts to aerobraking systems [2].

The potential market for a LEO-based OD solution consists in various types of missions, ranging from servicing commercial Earth satellites to supporting GEO and lunar missions. An orbiting depot placed in L1, on the other side, would primarily service customers looking to exploit cislunar (and beyond) development [2]. Since the placement of an OD is highly dependent on the missions to be supported and eventual clients' requirements, a comprehensive scenario could also consider both solutions, i.e. LEO and L1 depots, working together. In a scenario foreseeing a single OD, however, a L1-placed OD represents the best option for the reasons discussed above and it is therefore selected for the analysis in the following part of the study.

### 3.3. Preliminary design

The selected distribution facility includes a depot in L1 served by a Moon shuttle. In order to define the depot's features, it is necessary to determine the main requirements of the design. They can be divided into three categories: (i) technical requirements for cryogenic propellant storage, (ii) requirements imposed by available launchers to position the assembled depot in L1, (iii) requirements imposed by the chosen vehicle to perform the depot refueling from the Moon surface. The design is conceived to reach a compromise between these requirements and to define the optimal configuration.

Technical requirements for cryogenic propellant storage mainly refer to thermal insulation, zero boil-off and fluid management and transfer. The features of existing tanks for cryogenic propellant are taken into account to define the properties of the depot and its subsystem. In particular, the selected thickness and materials of LOX/LH2 tanks are in accordance with the values of Centaur's tanks (a balloon-like tanks developed by ULA) [9,13–17]. In this study, an optimal solution between the structure's thickness and the internal pressure at launch is adopted. This type of concept, in fact, allows a lighter structure via thickness reduction given that the necessary pressurization of the propellant is maintained at launch and while in service so to grant the required structural rigidity and avoid structural instabilities [13]. For

**Table 5**

Refueling costs from the Moon surface.

| Case 1: Refueling costs from the Moon surface          |                   |
|--------------------------------------------------------|-------------------|
| Refueling vehicle [20]                                 | CSDC Moon shuttle |
| Refueling vehicle payload [t] [20]                     | 25                |
| Estimated propellant for maneuver [t]                  | 50.02             |
| Propellant cost on Moon surface [\$/kg] [2]            | 500               |
| Estimated refueling cost (propellant + maneuver) [M\$] | 37.51             |
| Operation cost [M\$/year] [22]                         | 35                |

**Table 6**

Refueling costs from the Earth surface.

| Case 2: Refueling costs from the Earth               |                                           |
|------------------------------------------------------|-------------------------------------------|
| Refueling vehicle [25]                               | Falcon Heavy Expendable + Propellant tank |
| Refueling vehicle payload [t]                        | 17.5                                      |
| Launch costs [\$/kg]                                 | 6,511.4                                   |
| Estimated refueling cost (propellant + launch) [M\$] | 227.9                                     |
| Operation cost [M\$/year] [26]                       | 42.38                                     |

these reasons, the propellant depot is envisioned to be equipped with a pressurization system and, thanks to the reduced structural mass granted by the adopted monocoque configuration, also partially filled with propellant at launch.

The dimensions of the depot are selected in compliance with the performances of available launchers and refueling vehicles. The analysis is carried out considering both pre-existing technologies and designed vehicles, currently in testing phase. There are different launchers that can reach L1 for OD placement, whereas the selection of the Moon shuttle is more limited. The chosen refueling vehicle is designed by Cislunar Space Development Company (CSDC) and it can transfer to the depot up to 25 t of propellant from the lunar surface [9]. As it is propelled by LOX/LH2, this vehicle can be refueled directly by the lunar plant. To be compatible with the Moon shuttle's payload capability, the depot's capacity was fixed at 25 t of propellant mass so that it can be completely refilled with a single trip. The chosen O/F ratio and the actual dimension of the depot are selected considering the launcher's requirements in terms of fairing allowance and payload capability for the L1 destination. The payload mass budget allowed by the launcher is also taken into account to verify whether a full depot can be launched from Earth.

For a preliminary investigation, SpaceX's Falcon Heavy has been considered [14]. Its maximum launch capability depends on the final destination: for the L1 option, the maximum launch capability is only 17.5 t [18]. This means that the depot cannot be launched completely full. As mentioned above, the chosen structure for the OD is a balloon-like tank: following the design choice adopted for the Centaur upper stage [19], inox austenitic steel 301 is considered as main constitutive material. A capacity of 25 t LOX/LH2 with a mixture ratio of 6 [5] is selected, resulting in dimensions ( $h = 5.79$  m,  $d = 4$  m) chosen to be compliant with Falcon Heavy's fairings. The features of the design are shown in Table 1.

### 3.4. Refueling scenarios

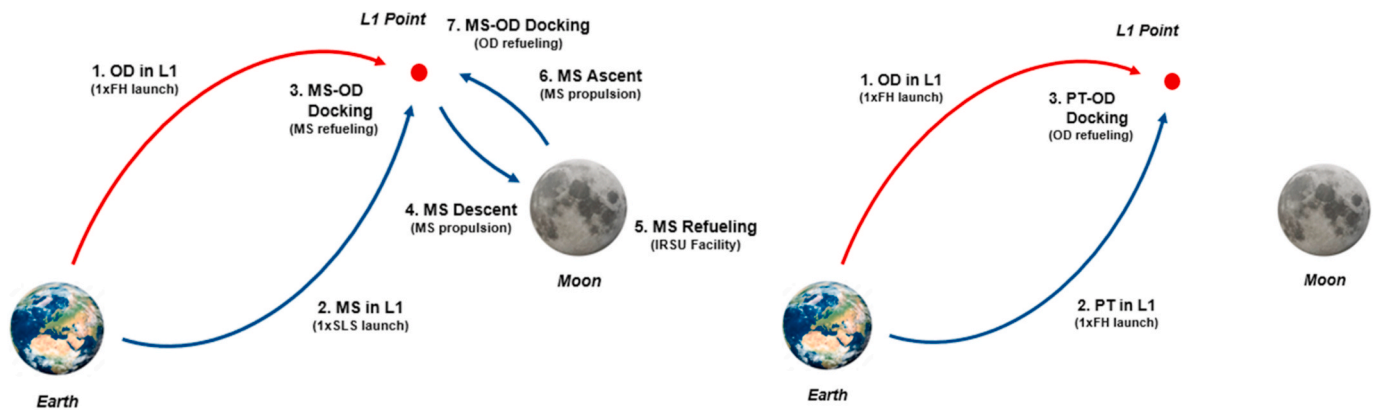
To evaluate the pros and cons of Moon-mined propellant availability and distribution, a comparison is made between two possible refueling scenarios for the L1 depot, namely, refueling performed from Earth and from the Moon-based ISRU facility. To clarify the exposition, we refer to the latter as Case 1 and to the former as Case 2.

#### 3.4.1. Vehicles

To refuel the depot with lunar propellant it is necessary to select a suitable vehicle which performs such maneuver. As mentioned above, the chosen vehicle is designed by CSDC and shown in Fig. 4 [20]. To date, no info about the lander reusability (i.e., how many times it can be refueled before maintenance is due) is made available by CSDC.

The data used for the analysis is shown in Table 2. Since the vehicle must be transferred to the Moon surface, the cost of its launch must also be considered. Because of its dimensions, the most suitable launcher for its transport is found to be NASA's SLS [21].

Being propelled by LOX/LH2, the Moon Shuttle is directly served by the lunar mining plant. The in-situ facility would provide the necessary



**Fig. 5.** Schematic representation of two refueling scenarios considered in the study. In Case 1 (left) the first 3 phases only take place once to deploy the Orbiting Depot (OD) and the Moon Shuttle (MS) at L1 point and perform a successive docking to allow the latter to embark the necessary fuel to carry out the descent on the Moon surface. Phases 4 to 7 are repeated to periodically refuel the OD. In Case 2 (right) only the first phase of OD deployment at L1 takes place once as Propellant Tanks (PT) must be periodically launched from Earth to refuel the OD.

propellant both for the transport and for the depot's refueling, with a total cost obtained by summing the two. Once placed in L1 point, the lander is intended to dock into the depot and refuel to perform the descent manoeuvre to the moon surface, making use of the fuel already present a launch in the OD. In this preliminary study, the propellant required by the lander to perform propulsive manoeuvres for its first docking into the depot is neglected. The costs due to the Design, Development, Testing and Evaluation (DDT&T) of both the fuel depot and the Moon shuttle are taken from literature [22] and are reported in Table 3.

The refueling vehicle needs to be transported on the Moon surface in order to perform refueling maneuvers. The costs of which are reported in Table 4. Because of its assumed dimensions [22], the only currently available launcher is NASA's SLS that is currently under testing [21].

### 3.4.2. Refueling maneuvers

The Moon-based refueling manoeuvres consists of three phases: (i) the Moon shuttle leaves the Moon surface with the maximum payload capability and performs an ascent maneuver; (ii) it then docks with the depot in L1 to allow the fuel transfer and (iii) it finally comes back landing with its only dry mass. For each phase, knowing the  $\Delta V$  requirements, the necessary propellant's mass for manoeuvre is estimated. The main assumption is that there are no propellant losses during the transfer; thus, the vehicle takes off with the sufficient propellant to perform such an operation. For any phase the Tsiolkovsky equation has been implemented to calculate the necessary propellant [23]. The cost breakdown is presented in Table 5.

The result of this study shows that the Moon shuttle can leave lunar surface, refuel L1 depot with 25 t of propellant and return to the Moon without needing to refuel, consuming about 50 t of LOX/LH2 propellant.

Refueling from Earth is then analyzed by considering an architecture compatible with the designed OD to be refilled. In this case, in the lift-off mass evaluation, the tank to be launched containing the propellant

needed to refill the OD consists only of the structure, the propulsion and the thermal management subsystems shown in Table 1. As already mentioned, two Falcon Heavy launches would be necessary to completely refuel the OD in this scenario. The total cost is estimated by summing the propellant's cost on Earth [24], that is negligible, and the launch costs. By considering that the total launch cost for a FH Expendable is of 150 M\$, the cost estimation has been made as follows: the maximum capability for FH Expendable is 17.5 t [25], but the requirement is to bring 26.69 t as 25 t of propellant are needed to refill the OD and the rest is necessary to transfer the refilling tank with its subsystems, as derived by Table 1. It means that  $\sim 1.5$  launches are ideally required to completely refuel the OD; the total cost is obtained by multiplying the real launch cost by the ideal number of launches, totaling the price of 227.9 M\$, as shown in Table 6.

A schematic representation of the various phases which constitute the Case 1 and Case 2 refueling scenarios is shown in Fig. 5.

### 3.5. Overall costs of the case studies

Tables 7 and 8 sum up the computation and the comparisons that have been performed in order to carry out the cost estimation for Moon- and Earth-based OD refueling. Refueling from Earth of course does not require a Moon-based refueling vehicle: all the associated costs, such as design, test evaluation, manufacturing and transportation of such vehicle therefore do not appear in the table. On the contrary, it is apparent that performing a refueling maneuver from Earth surface basically translates into operating with a quasi-heavy lift approach, implying multiple required launches to refill the OD as the cheapest available launcher, i.e., Falcon Heavy Expendable, does not match the maximum capability of the depot. Therefore, the launch, the operations, the propellant's transfer and all the other maneuvers must be performed more than one time, leading to a more difficult and less reliable refueling strategy.

**Table 7**

Costs associated to L1-based OD refueling from the Moon (\* per year).

| Case 1: Refueling from Moon |                |                               |       |
|-----------------------------|----------------|-------------------------------|-------|
| Fixed costs                 |                | Parameter                     | [M\$] |
|                             | DDT&E          | Depot                         | 400   |
|                             |                | Moon Shuttle                  | 1,200 |
|                             | Manufacturing  | Depot                         | 25    |
|                             |                | Moon Shuttle                  | 700   |
|                             | Transportation | Depot in L1                   | 150   |
|                             |                | Moon Shuttle on Lunar surface | 670   |
| Recurring costs             | Refueling      | 37.51                         |       |
|                             | Operations     | 35*                           |       |

**Table 8**

Costs associated to L1-based OD refueling from the Earth surface (\* per year).

| Case 2: Refueling from Earth |                |             |        |
|------------------------------|----------------|-------------|--------|
| Fixed costs                  |                | Parameter   | [M\$]  |
|                              | DDT&E          | Depot       | 400    |
|                              | Manufacturing  | Depot       | 25     |
|                              | Transportation | Depot in L1 | 150    |
| Recurring costs              | Refueling      |             | 227.9  |
|                              | Operations     |             | 42.38* |

**Table 9**

Data input assumptions for the project: from Moon to L1 and the project From Earth to L1.

|                                      |                                          | Case 1: From Moon to L1 | Case 2: From Earth to L1 |
|--------------------------------------|------------------------------------------|-------------------------|--------------------------|
| Revenues<br>(Operating phase)        | Final fuel price [M\$/Mt]                | 5.07                    | 10.55                    |
|                                      | Yearly demand of fuel [Mt]               | 280                     | 280                      |
|                                      | Total yearly revenues [M\$]              | 1,419                   | 2,954                    |
| Operating Costs<br>(Operating phase) | Refueling from moon per journey [M\$]    | 37.51                   | 227.90                   |
|                                      | Yearly missions from moon to depot [M\$] | 420.11                  | 2,552.48                 |
|                                      | Yearly operation costs [M\$]             | 35                      | 42.38                    |
|                                      | Total yearly operating costs [M\$]       | 455.11                  | 2,594.86                 |
| Investment Costs                     | DDT&E total costs [M\$]                  | 1,600                   | 400                      |
|                                      | Manufacturing costs [M\$]                | 725                     | 25                       |
|                                      | Launch costs [M\$]                       | 820                     | 150                      |
|                                      | Total capital expenditure (CAPEX) [M\$]  | 3,145                   | 575                      |
| Depreciation & Amortization          | Depreciation rate                        | 10.00%                  | 10.00%                   |
|                                      | Yearly depreciation [M\$]                | 314.5                   | 57.5                     |
| Tax Rate                             | Corporate tax rate                       | 28.00%                  | 28.00%                   |
| Weighted Average Cost of Capital     | WACC discount rate                       | 13%                     | 11.50%                   |
|                                      | NPV                                      | 0                       | 0                        |

## 4. Business model

### 4.1. Economic assumptions

The assessment of the economic sustainability of the two refueling models (Case 1 and Case 2) requires a comparison between the two alternatives. Our aim is to identify the existence of a competitive space that allows an economic choice between the two cases and requires estimations of the expected future cash flows. The achievement of this goal is possible by forecasting future cash flows of the firms involved in the two cases. In our study we assume the presence of a single transporter/distributor; we justify this assumption by that considering the required investment is too large for multiple companies to enter the market together. This assumption simplifies, at this stage of our study, the analysis of the competition in the market. Once estimated the expected future cash flows, the rationale of the NPV criterion is the basis to verify the presence of a competitive space based on the identification of two minimum prices for the final users in every context, i.e., in Case 1 and Case 2, respectively. These estimations, requiring a discounting process, are linked to consistent assumptions in terms of riskiness and related opportunity cost of capital.

The Net Present Value formula is here presented:

$$NPV_0 = - \sum_{t=0}^N \frac{CAPEX_t}{(1+r)^t} + \sum_{t=N+1}^T \frac{CF_t}{(1+r)^t}$$

where  $CF_t$  are the net cash inflow-outflow,  $r$  is the discount rate, and  $T$  is the number of years of the project for which the cash flows are estimated and discounted.  $CAPEX_t$  are the cash outflows for the construction phase of the analyzed projects. The fundamental principle of value creation is that the project creates value by investing the capital raised from investors to generate cash flows at rates of return exceeding the cost of capital (the rate that investors require for bearing the risk associated with the uncertainties of the projected cash flows). Both in Case 1 and Case 2,  $N = 5$  and  $T = 10$ : the other components of the NPV formula will be described in the following sections.

#### 4.1.1. General assumptions

Our model requires some hypotheses that are common to the two scenarios. In terms of rationale, the valuation requires estimating demand, recurring costs, set-up costs, and cost of capital for the two cases. All the assumptions are specific for each alternative but the quantity of the supplied propellant, that is constant and equal to 280 Mt per year [2]. This implies that at least eleven journeys per year are necessary to recharge the depot (capability 25 Mt).

The estimations about the costs per each case have been already described above in Tables 7–8. Despite the assumptions in terms of Demand Quantity are available, the price that allows the revenues estimation is the unknown variable in our Cash flow estimation and the real aim of this study. The final output of our research that generates the cash flow estimation prospectus is presented in Table 9, where the first row (“Final Price Fuel”) for the two Cases is the final output allowing us to understand the competitive distance between the two choices.

Considering that one of our goals is to understand if there is room for a commercial market to emerge, our interest is in the minimum selling price that it is necessary to set for the “Case 1” and the “Case 2”. At this price, the Net Present Value is equal to zero, implying that a lower price destroys the convenience for the operator.

The market belongs to the operator who can sell the propellant at the lower price and, additionally, the market exists between the two boundaries. If our study confirms a difference between the two minimum prices, the potential market could incentivize the intervention of public actors, willing to provide initial grants to stimulate the participation of private firms. This intervention lowers the amount of investment required to private companies and reduces their risk exposure.

Uncertainty is one of the most relevant aspects in the valuation of every investment project, but it is crucial in innovative initiatives. The most peculiar aspect of the two alternatives here analyzed is, likely, the total absence of a similar work fruit of the human ingenuity. This is particularly true for Case 1. In accordance with this consideration, we estimate the two risk and expected return profiles, the first one suited to the “from earth refueling” alternative and the latter one elaborated for the “from moon refueling” project. In this part of the paper, we do not model the risk (the following part presents the Scenario and Montecarlo analysis), but we consider its presence; this requires a cost of capital estimation different from the risk-free rate.

#### 4.1.2. Specific assumption: case 2 – cost of capital

Even though there is a non-negligible difference between the aerospace projects already carried out in human history and the project object of this study, it is possible to identify a portfolio of possible comparable companies among the constituents of the S&P Kensho Space Index, a Stock Index dedicated to companies focused on the space industry. Starting with their parameter of risk remunerated by the financial market, identified, according to the literature and the operating practice, with the statistical Beta, we initially assume a similar financing mix in terms of Debt and Equity contribution to our project. It is remarkable that according with our expectations, the risk parameter for our portfolio of peers is permanently high, around 1.70, representing high sensitivity to the volatility of the overall market. We precise that, at the birth of these innovative projects, we are using the Beta concept and the assumptions in terms of financial structure with a long-time perspective, as a possible measure of equilibrium solution, suitable for our purposes.

Financial procedures for the treatment of risk allow us to recalculate a beta for our project, based on a similar financial structure [27]. We use the Bloomberg database as a basis for our calculations and are able to estimate a final cost of capital aligned with the average Weighted Average Cost of Capital (WACC), the discount rate for the expected cash flows characterizing the portfolio of comparable companies (namely, the WACC for our discounting process is equal to 11.50%). This discount rate, joined with the assumption of a demand equal to 280 Mt allows us to find out the minimum price (10.55 M\$/Mt) that keeps the

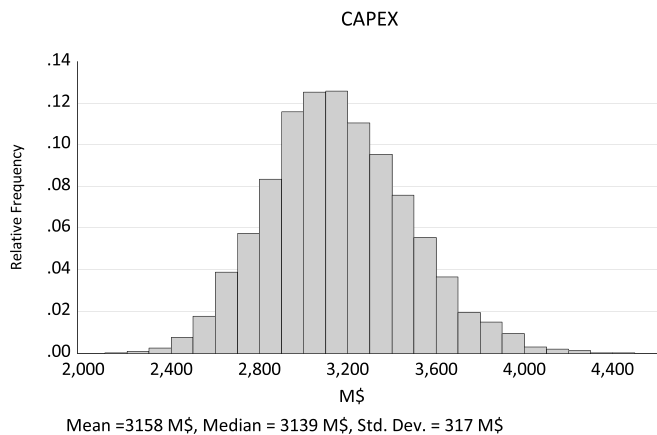


Fig. 6. The lognormal distributions of the CAPEX of the Moon firm.

convenience of the project.<sup>1</sup>

We highlight that we unconventionally explained the “Case 2 Cost of capital estimation” before presenting “Case 1 Cost of Capital” only for expository clarity due to the better comparability with existing activities.

#### 4.1.3. Specific assumption – case 1 – cost of capital

When considering the initiative of recharging the Depot in L1 with the propellant mined on the lunar surface, we see not only a notable increase in the necessary funds (the moon shuttle is the most relevant incremental outflow) but also an exacerbation of the perceived risk. No kind of human presence is currently active on the lunar territory, making it extremely challenging to identify a group of comparable companies for the cost of capital. The parameter used to explain the increase in risk in the new context is linked to the size of the necessary investment. The size has already demonstrated to be a good predictor of the increase in risk, read as probability of obtaining lower than expected annualized returns on investments, so we decide to link the increase in risk to this marginal effect [28]. The new beta is therefore 13% higher than the beta for project “from Earth” and the cost of financial sources rises accordingly to this assumption.

Carrying out the necessary calculations, we estimate a final WACC discount rate for this case, equal to 13%. Even here, we identify the minimum price that keeps the NPV convenient, i.e., not below zero: in this case the price for fuel has to be set up at 5.07 M\$/Mt.

This result is particularly important to influence the choice between the two projects: refueling from the Moon, although more expensive during the construction phase and riskier, appears more affordable in terms of final price and makes the development of a commercial market, possibly supported by public intervention, more likely and interesting for the stakeholders. Establishing the minimum price that allows a non-negative net present value for the two projects we find can the price range for Case 1 that makes this alternative attractive: 10.55 M\$/Mt – 5.07 M\$/Mt. Perhaps even more interesting, we can state that Case 1 is a competitive choice even at a final price of 10.55 M\$/Mt. At this last possible price, i.e., 10.55 M\$/Mt, the Case 1 NPV results in about 3 \$billion. The correspondent Internal Rate of Return (IRR), at that price, is 16%, higher than the cost of capital.

An interesting conclusion is linked to the risk considerations. The operative risk is different between the two alternatives (Case 1 and Case 2). This risk can affect the financial sustainability of the investment, but the governmental intervention could mitigate the volatility, in particular its downside, opening to the possibility of a Public-Private

<sup>1</sup> We will refer to 10.55 M\$/Mt as the minimum price for Case 2. Similarly, 5.07 M\$/Mt as the minimum price for Case 1.

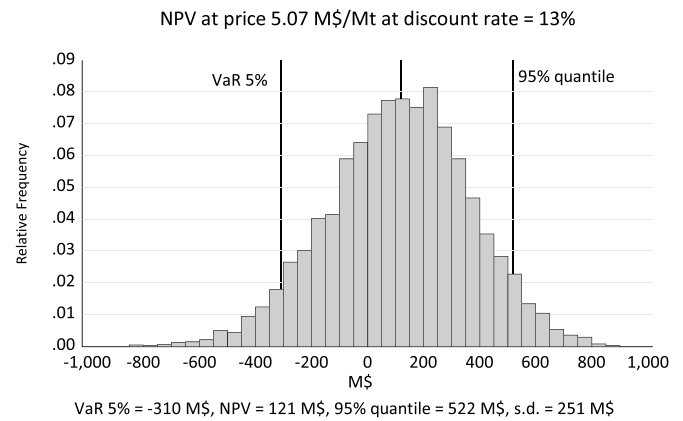


Fig. 7. The histograms of the NPV Case 1.

Partnership (PPP) model designed as in the more traditional infra-structural Project Financing.

#### 4.1.4. Business model summary

The models based on the expected values (according to our estimates) are computed using the estimated capital costs (Section 4.1.2 and 4.1.3), fixed and recurring costs (Tables 7 and 8), and on the volume (280 Mt) and price (5.07 and 10.55 M\$/kg) of propellant sold each year for each case. The business model, with the volatility in inputs and outputs included in the estimated cost of capital, is discussed above, and is summarized in Table 9 for Case 1 and Case 2. But the model lacks the explicit mention of the risk of under or overspending over each cost item, or of the risk of changing the fuel prices through time.

## 5. Scenario and Monte Carlo analysis

In this section, we do a scenario and Monte Carlo analysis to gain further insights on the profitability and risk of the project. Specifically, we assume that the costs and the prices are random lognormal variables, following standard practices. We highlight that a lognormal distribution is particularly suitable to describe economic variables, in particular to account possible extra high values of revenues and expenses.

A random variable,  $X$ , has a lognormal distribution, with parameters  $\mu$  and  $d$ , if  $\ln(X)$  is normally distributed with mean  $\mu$  and standard deviation  $d$ . The variable  $X$  is always positive, right skewed, and its values are massed around the Median, the value  $\exp(\mu)$ . Considering Table 9, the variables to which we assign a distribution are: final fuel price, total yearly operating costs, and CAPEX. For both cases, we consider the value in Table 9 as the median of the lognormal distribution and a standard deviation of 10% of the median.

For example in the case of CAPEX for Case 1 we assume  $\exp(\mu) =$

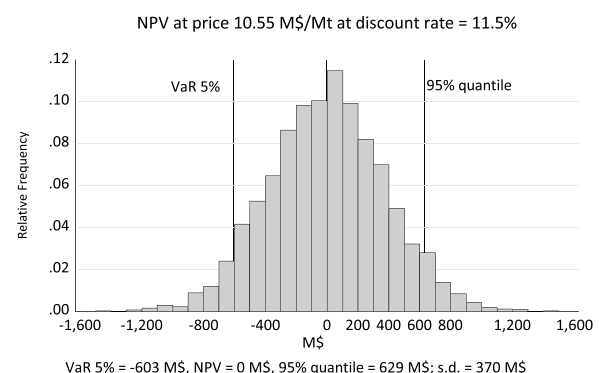


Fig. 8. The histogram of the NPV Case 2.

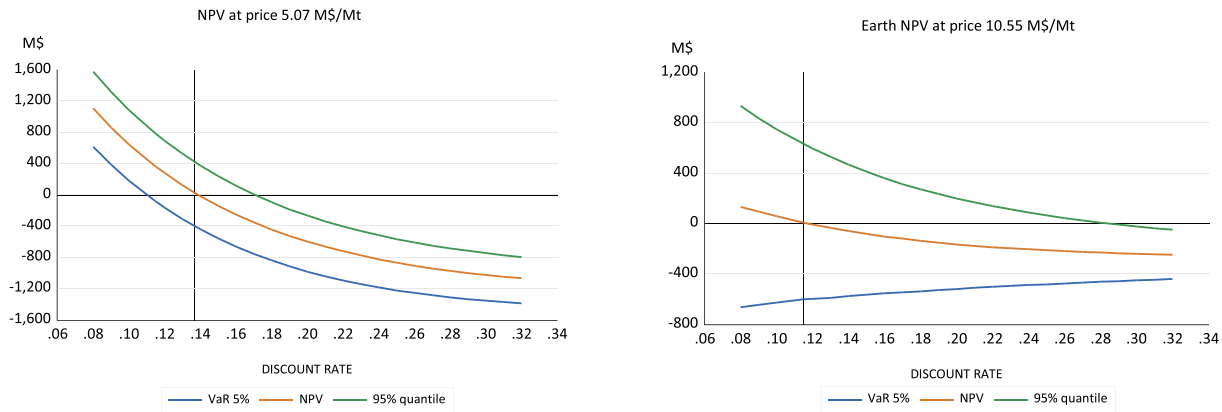


Fig. 9. NPV as a function of the discount rates.

3,145 M\$ (see in Table 9), so a random value for  $CAPEX_i^M$  is specified as:

$$CAPEX_i^M = \exp(\mu) * u_i \rightarrow \ln(CAPEX_i^M) = \mu + e_i; e_i \sim N(0, d^2)$$

We show in Fig. 6 the histogram of the simulated CAPEX lognormal distribution of Case 1. We forecast a safe scenario – low risk, so that the standard deviation of each random variable is  $d = 10\%$  of the median. In the simulation of CAPEX for case 1 we obtain a standard deviation of 314.

As presented in Equation (1), cash flows for the Moon and Earth firms have two parts. The first that lasts five years is the construction phase, where the initial Capital (Depot, in Case 2; Depot and Moon Shuttle in Case 1) is built, and the operating phase lasts the following 10 years.

In the simulation exercises, we compute the NPV from the expected future cash flows 5,000 times for both Case 1 and Case 2. At each iteration, the random values are sampled from the lognormal distributions. The 5,000 solved NPVs are represented in Figs. 7 and 8 for the two Cases.

Remember that in the “Expected” NPV calculations, the prices have been fixed to get NPV zero, (the minimum price in Case 1 was fixed at 5.07 M\$/Mt with a WACC discount rate at 13%, while for Case 2 it was set at 10.55 M\$/Mt with an 11.5% WACC discount rate).

The Moon and Earth NPV graphs show different structures. The “Expected” NPV computed from the values in Table 9, is zero and coincides with the median in histograms in Figs. 7 and 8, so the probability of negative NPVs is 50%, the same for both cases. For Case 1 the simulated values around the “Expected” NVP fall in the 90% confidence interval, between –400 M\$ to 380 M\$. For Case 2 this interval is larger, –604 M\$ to 629 M\$, and the random NPV has a higher standard deviation (370 M\$ versus 250 M\$), implying that the risk is larger for Case 2 than for Case 1. It is interesting to note that most of the risk of the NPV of Earth firms is attached to the yearly missions, while most of the risk of Moon firms NPV comes from the initial CAPEX.

We can also simulate the NPVs as functions of the discount rate (see Fig. 9). These figures consider the minimum prices (for a null NPV at the WACC in Table 9) as the discount rate changes, the central line in the graphs is the “Expected” NVP, the lower line connects the values of the 5% quantile of the simulated NVP (VaR 5%), the higher line connects the values of the 95% quantile of the simulated NVP (95% quantile). Look at the differences of the costs between the Moon and the Earth projects in Table 9: the increase of the discount rate lowers the Moon NPV more than the Earth NPV, as the higher rate discounts the high Operating costs of the Earth transporter, less the CAPEX of the Moon transporter.

In the last figure, Fig. 10, the Case 1 NPV is simulated assuming that the median of the lognormal price is 10.55 M\$/Mt, which doubles the Case 1 minimum price, but it is still competitive with Earth refueling firms. This almost triples the NVP at each discount rate, generates an IRR of circa 32% and, for example, the probability of high profit and negative NPV is zero at a discount rate 20%.

We highlight the relevance of the result emerging from the

comparison of Figs. 9 and 10: the NPV generated by Case 1 can afford a higher level of risk and is positive until around a 32% discount rate. This level is also the maximum cost of capital this project can afford, implying a better appeal for private equity investors and allowing the definition of a more sophisticated financial structure. This implies that the project can tolerate a higher level of debt because the recourse to borrowed funds increases the risk and the expected return on a project [29], as the trend followed by the Capital Market Line indicates. The increase in the expected return is the result of the additional debt, but, on the other hand, the marginal unit of debt does not generate negative effects on the value immediately, but only when its usage exceeds the optimal level that equalizes benefits and costs it generates. Trade-off theories analyze this aspect in detail. About the effects of debt introduction on value, see also [30,31].

As a result, a higher level of debt tolerance can make the project suitable to be financed with project finance loans by banks or with funds provided by institutional investors because close to the risk and return goals defined by these financial actors.

## 6. Conclusions

Our study compares two possible projects to refuel an on-orbit depot of propellant available for in space recharging activities from the Moon or Earth in order to verify two separate conclusions. We verify that operating an on-orbit depot is feasible and that recharging it is economically more affordable from the Moon as its price to final customers, which guarantees the profitability of the project, is lower than recharging the depot from Earth, thus making this option more suitable

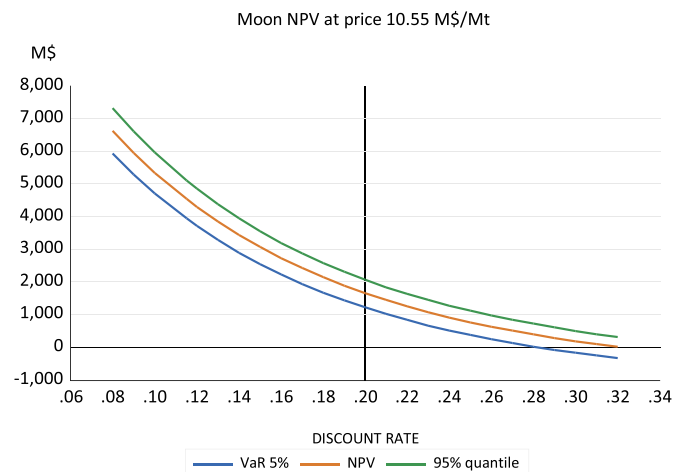


Fig. 10. Moon NPV at the Earth minimum's price.

to facilitate the emergence of a private demand, together with saving costs to public space exploration missions. In addition, the more expensive alternative from Earth is also more exposed to volatility implying higher levels of the underlying operational risks.

Besides, the price gap suggests that there is enough room for more than an operator that transports and distributes propellant from the lunar surface, making it reasonable for governments to disburse initial funding to multiple ventures, at least to support technology demonstration and validation phases. In addition, the risk and return expectations are not only attractive to private equity investors but they are also potentially suitable to balance the financial structure of the venture with more risk-adverse financiers such as institutional investors.

In conclusion, our study lays the foundation for approaching the development of transportation and distribution infrastructures for lunar-based propellant and it suggests that there are the conditions for enhancing private capital formation with which governments can share the risks of the venture as occurs in common infrastructure financing.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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